INTEGRALS DEFINITE

1.	$\int_0^{\frac{\pi}{6}} \sec^2(x - \frac{\pi}{6})\, dx$ is equal to : (a) $\frac{1}{\sqrt{3}}$ (b) $-\frac{1}{\sqrt{3}}$ (c) $\sqrt{3}$ (d) $-\sqrt{3}$
2.	Evaluate : $\int_0^{\frac{\pi}{2}} [\log(\sin x) - \log(2\cos x)]\, dx.$
3.	Evaluate : $\int_0^{\frac{\pi}{2}} e^x \sin x\, dx$
4.	Evaluate $\int_{\log\sqrt{2}}^{\log\sqrt{3}} \frac{1}{(e^x + e^{-x})(e^x - e^{-x})}\, dx$
5.	Evaluate $\int_{-1}^{1} \lvert x^4 - x \rvert\, dx.$
6.	The value of $\int_0^{\pi/2} \log \tan x\, dx$ is : (a) $\frac{\pi}{2}$ (b) 0 (c) $-\frac{\pi}{2}$ (d) 1

7.	Evaluate : $$\int_0^{\pi/4} \log(1+\tan x)\,dx$$
8.	Find : $$\int_1^4 \frac{1}{\sqrt{2x+1}-\sqrt{2x-1}}\,dx$$
9.	If $\int_0^a 3x^2\,dx = 8$, then the value of 'a' is : (a) 2 (b) 4 (c) 8 (d) 10
10.	Evaluate : $$\int_{\pi/4}^{\pi/2} e^{2x}\left(\frac{1-\sin 2x}{1-\cos 2x}\right)dx$$
11.	Evaluate : $$\int_{-2}^{2} \frac{x^2}{1+5^x}\,dx$$
12.	Evaluate : $$\int_0^{\pi/2} \sqrt{\sin x}\cos^5 x\,dx$$

13.	The value of $\int_{0}^{\frac{\pi}{4}} (\sin 2x)\, dx$ is (a) 0 (b) 1 (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$
14.	**Assertion (A) :** $\int_{2}^{8} \frac{\sqrt{10-x}}{\sqrt{x}+\sqrt{10-x}}\, dx = 3$ **Reason (R) :** $\int_{a}^{b} f(x)\, dx = \int_{a}^{b} f(a+b-x)\, dx$
15.	Evaluate : $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{\cos 2x}{1+\cos 2x}\, dx$
16.	. Evaluate : $\int_{0}^{\pi} \frac{x}{1+\sin x}\, dx$
17.	$\int_{-\pi/4}^{\pi/4} x^3 \cos^2 x\, dx$ is equal to : (A) 0 (B) -1 (C) 1 (D) 2
18.	Evaluate : $\int_{-4}^{0} \lvert x+2 \rvert\, dx$

19.	The value of $\int_0^{\pi/6} \sin 3x \, dx$ is : (a) $-\frac{\sqrt{3}}{2}$ (b) $-\frac{1}{3}$ (c) $\frac{\sqrt{3}}{2}$ (d) $\frac{1}{3}$
20.	Evaluate : $\int_0^{\pi/2} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$
21.	Evaluate : $\int_1^3 (\lvert x-1 \rvert + \lvert x-2 \rvert)\, dx$
22.	$\int_a^b f(x)\, dx$ is equal to : (A) $\int_a^b f(a-x)\, dx$ (B) $\int_a^b f(a+b-x)\, dx$ (C) $\int_a^b f(x-(a+b))\, dx$ (D) $\int_a^b f((a-x)+(b-x))\, dx$
23.	Evaluate : $\int_0^{\frac{\pi^2}{4}} \frac{\sin\sqrt{x}}{\sqrt{x}} dx$

24.	Evaluate : $$\int_1^3 (\lvert x-1\rvert + \lvert x-2\rvert + \lvert x-3\rvert)\, dx$$
25.	Evaluate : $$\int_0^{\frac{\pi}{4}} \frac{\sin x + \cos x}{9 + 16 \sin 2x}\, dx$$
26.	Evaluate : $$\int_0^{\frac{\pi}{2}} \sin 2x \tan^{-1} (\sin x)\, dx$$
27.	If $\int\limits_0^2 2\, e^{2x}\, dx = \int\limits_0^a e^x\, dx$, the value of 'a' is : (A) 1 (B) 2 (C) 4 (D) $\frac{1}{2}$
28.	Evaluate : $$\int_0^{a^3} \frac{x^2}{x^6 + a^6}\, dx$$

29.	Evaluate : $$\int_{-2}^{2} \sqrt{\frac{2-x}{2+x}}\, dx$$
30.	$\int_{-a}^{a} f(x)\, dx = 0$, if : (A) $f(-x) = f(x)$ (B) $f(-x) = -f(x)$ (C) $f(a-x) = f(x)$ (D) $f(a-x) = -f(x)$
31.	Evaluate : $$\int_{0}^{\pi/2} \sin 2x \cos 3x\, dx$$
32.	Evaluate : $$\int_{0}^{\pi/4} \frac{1}{\sin x + \cos x}\, dx$$

33.	The value of $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$ is : (A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{18}$
34.	) Evaluate : $\int_0^{\frac{\pi}{4}} \frac{x\,dx}{1+\cos 2x+\sin 2x}$
35.	The value of $\int_{-1}^{1} x\,\lvert x\rvert\,dx$ is : (A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $-\frac{1}{6}$ (D) 0
36.	Evaluate : $\int_0^{\pi} \frac{e^{\cos x}}{e^{\cos x}+e^{-\cos x}}\,dx$
37.	$\int_{-1}^{1} \frac{\lvert x\rvert}{x}\,dx,\ x \neq 0$ is equal to (A) -1 (B) 0 (C) 1 (D) 2

38.	Evaluate : $\displaystyle\int_0^{\frac{\pi}{4}} \frac{dx}{\cos^3 x\sqrt{2\sin 2x}}$
39.	Evaluate : $\displaystyle\int_0^{\pi} \frac{\sin 2px}{\sin x}\,dx$, $p \in N$.
40.	f and g are continuous functions on interval [a, b]. Given that $f(a - x) = f(x)$ and $g(x) + g(a - x) = a$, show that $\displaystyle\int_0^a f(x)\,g(x)\,dx = \frac{a}{2}\int_0^a f(x)\,dx$.
41.	The value of $\displaystyle\int_0^1 \frac{dx}{e^x + e^{-x}}$ is : (A) $-\frac{\pi}{4}$ (B) $\frac{\pi}{4}$ (C) $\tan^{-1} e - \frac{\pi}{4}$ (D) $\tan^{-1} e$
42.	Evaluate : $\displaystyle\int_0^{3/2} \lvert x\cos \pi x \rvert\,dx$
43.	$\displaystyle\int_0^{\pi/2} \cos x . e^{\sin x}\,dx$ is equal to : (A) 0 (B) $1 - e$ (C) $e - 1$ (D) e

44.	Evaluate : $$\int_1^4 \left(\lvert x-2\rvert+\lvert x-4\rvert\right)\,dx$$
45.	Evaluate : $$\int_0^{\pi/2} \frac{x}{\sin x+\cos x}\,dx$$
46.	For a function f(x), which of the following holds true ? (A) $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$ (B) $\int_{-a}^{a} f(x)\,dx = 0$, if f is an even function (C) $\int_{-a}^{a} f(x)\,dx = 2\int_0^a f(x)\,dx$, if f is an odd function (D) $\int_0^{2a} f(x)\,dx = \int_0^a f(x)\,dx - \int_0^a f(2a+x)\,dx$
47.	Evaluate : $$\int_0^{\pi/2} \frac{x}{\cos x+\sin x}\,dx$$